

Title Module – 2

Autonomic Pharmacology

LS 2.3 - ANS Regulation of Different Functions of the Body

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ANS REGULATION OF DIFFERENT FUNCTIONS IN THE BODY

LS – 2.3: LEARNING OBJECTIVES:

- To describe the effects mediated by the autonomic nervous system
- To classify the receptors that mediate the effects of the ANS and describe their signal transduction mechanisms
- To describe how drugs influence the autonomic nervous system

ANS REGULATION OF DIFFERENT FUNCTIONS IN THE BODY

INTRODUCTION:

ANS regulates functions of various organs and tissues to optimize response to internal and external stimuli

Parasympathetic actions: Mediated through muscarinic receptors

Sympathetic actions: Mediated through adrenergic receptors

Drugs modify ANS activity by:

- Increasing activity (agonists)

- Decreasing activity (antagonists)

PARASYMPATHETIC NERVOUS SYSTEM

Effector Organ	Receptor Subtype	Response to Cholinergic Stimulation
Eye:		
Sphincter muscle, iris	M ₃	Contract to cause miosis
Ciliary muscle	M ₃	Contract for near vision
Heart:		
Sinoatrial node	M ₂	Decrease rate of depolarization
Atria	M ₂	Decrease contractility
Atrioventricular node	M ₂	Decrease rate of depolarization
Arteries, arterioles:		
Endothelial cells	M ₁ , M ₃	Vasodilation (predominant) – mediated by nitric oxide (via cGMP)
Smooth muscle cells	M ₁ , M ₃	Vasoconstriction by direct action on smooth muscle

PARASYMPATHETIC NERVOUS SYSTEM ... CONT'D

Effector Organ	Receptor Subtype	Response to Cholinergic Stimulation
Respiratory:		
Tracheal, bronchial smooth muscle	M ₁ , M ₃	Constriction
Bronchial glands	M ₂ , M ₃	Increased mucous secretion
Gastrointestinal tract:	M ₂ , M ₃	Increased contractility, motility and secretions
Urinary Bladder:		
Detrusor muscle	M ₂ , M ₃	Contraction
Trigone & sphincter muscle	M ₂ , M ₃	Relaxation
Salivary glands:	M ₂ , M ₃	Increased watery secretion
Erectile tissue:	M ₂ , M ₃	Vasodilation of cavernosa of penis and clitoris, increased blood pooling

MUSCARINIC RECEPTORS AND THEIR EFFECTOR SYSTEMS

Cholinergic Receptor subtype	Receptor distribution	Signal transduction activated
Muscarinic, M ₁	CNS neurons, smooth muscle	G _q : ↑ IP ₃ and DAG → reduced K ⁺ conductance (depolarization, excitation)
Muscarinic, M ₂	Cardiac conduction system, myocardium	G _i : Inhibit adenylyl cyclase → ↓ cAMP → ↓ Ca ²⁺ conductance, ↑ K ⁺ conductance (inhibition)
	Presynaptic nerve endings	G _i : Inhibit adenylyl cyclase → ↓ cAMP → ↑ K ⁺ conductance (decrease neurotransmission)
Muscarinic, M ₃	Exocrine glands, smooth muscle	G _q : ↑ IP ₃ → increase intracellular calcium (stimulation)
Muscarinic, M ₄	CNS	G _i : Inhibit adenylyl cyclase → ↓ cAMP (inhibition)
Muscarinic, M ₅	CNS: very localised expression in substantia nigra, salivary glands Iris/ciliary muscle	G _q : ↑ IP ₃ (excitation)

SYMPATHETIC NERVOUS SYSTEM

Effector Organ	Receptor Subtype	Response to Adrenergic stimuli
Eye:		
Pupillary dilator muscle	α_1	Contracts to cause mydriasis
Ciliary muscle	β_2	Relaxes to allow for far vision
Heart:		
Sinoatrial node	β_1, β_2	Increased rate of depolarization (increased heart rate - positive chronotropy)
Atrioventricular node	β_1	Increased conduction rate
Atria, ventricles	β_1	Increased contractility (positive inotropy)
Arteries, arterioles:		
Arterial smooth muscle cells in most vascular beds	α_1, α_2	Contraction (vasoconstriction)
Arteries/arterioles in skeletal muscle and brain	β_2	Relaxation (vasodilatation)
Respiratory tract:		
Tracheal, bronchial smooth muscle	β_2	Relaxation
Bronchial glands	α, β_2	Decreased mucous secretion

SYMPATHETIC NERVOUS SYSTEM ... CONT'D

Effector Organ	Receptor Subtype	Response to Adrenergic stimulation
Gastrointestinal tract:		
Smooth muscle wall	β_2	Relaxation, decreased motility
Sphincters	α_1	Contraction, decreased transit time
Urinary Bladder:		
Detrusor muscle	β_2	Relaxation, decreases voiding
Trigone & sphincter muscle	α_1	Contraction, decreases voiding
Metabolic function:		
Liver	α_1, β_2	Glycogenolysis
Fat cells	β_3	Increased lipolysis
Kidneys	β_1	Renin release
Sexual organs:	α_1	Ejaculation, orgasm
Adrenal Medulla:	N_N	Epinephrine release

ADRENERGIC RECEPTORS AND THEIR EFFECTOR SYSTEMS

Adrenergic receptor	G-protein coupling	Signal transduction processes
β_1	G_s	Activate adenylate cyclase $\rightarrow \uparrow$ cAMP $\rightarrow$ open Ca^{2+} channels $\rightarrow \uparrow$ intracellular Ca^{2+} levels
β_2	G_s	Activate adenylate cyclase $\rightarrow \uparrow$ cAMP $\rightarrow \uparrow$ OR $\downarrow$ intracellular Ca^{2+} levels
β_3	G_s	Activate adenylate cyclase $\rightarrow \uparrow$ cAMP $\rightarrow$ activate fatty acid lipase
α_1	G_q	Activate phospholipase C, D and $A_2 \rightarrow$ open Ca^{2+} channels $\rightarrow \uparrow$ intracellular Ca^{2+} levels
α_2	$G_{i,o}$	Decrease adenylate cyclase $\rightarrow \downarrow$ cAMP $\rightarrow$ open K^+ channels $\rightarrow$ close Ca^{2+} channels $\rightarrow \downarrow$ intracellular Ca^{2+} Activate phospholipase C and $A_2 \rightarrow$ open Ca^{2+} channels $\rightarrow \uparrow$ intracellular Ca^{2+}

HOW DRUGS INFLUENCE THE AUTONOMIC NERVOUS SYSTEM

- **Receptor Agonists:**

- Mimic the effects of endogenous neurotransmitters
- Activate receptors
- Examples: Pilocarpine (muscarinic agonist), Adrenaline (adrenergic agonist)

- **Receptor Antagonists:**

- Block the effects of endogenous neurotransmitters
- Prevent receptor activation
- Examples: Atropine (muscarinic antagonist), Propranolol (β -blocker)

Clinical Examples of Drug Effects

- **Atropine:** Blocks parasympathetic effects (tachycardia, dry mouth, mydriasis)
- **Neostigmine:** Inhibits AChE → increases ACh → treats myasthenia gravis
- **Tricyclic antidepressants:** Inhibit NA reuptake → prolong NA effects
- **α -agonists (phenylephrine):** Nasal decongestants
- **β -blockers (propranolol):** Reduce heart rate, treat hypertension

CLASSIFICATION OF DRUGS AFFECTING THE AUTONOMIC NERVOUS SYSTEM

Parasympathetic nervous system

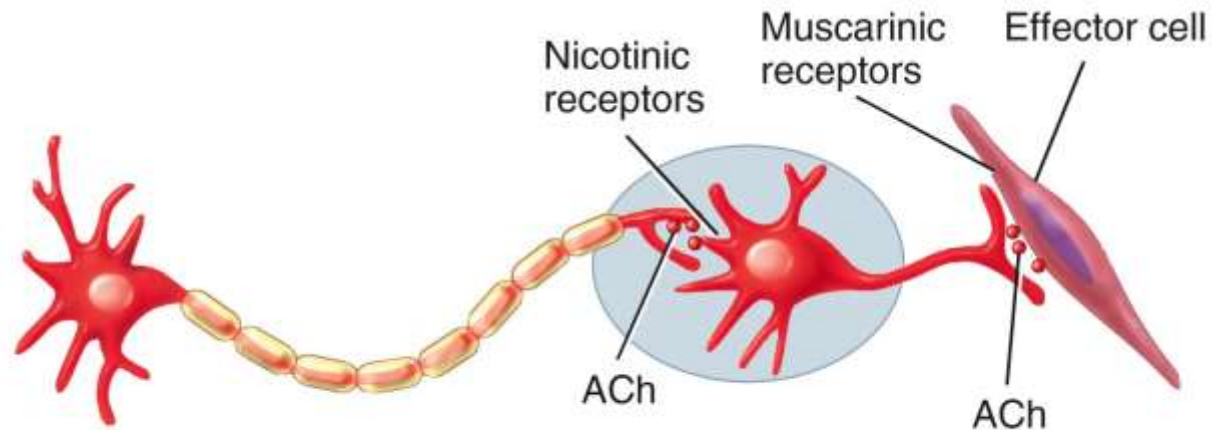
- Mimic acetylcholine (on muscarinic receptors) = muscarinic agonists = **Parasympathomimetic**
- Block acetylcholine (on muscarinic receptors) = anti-muscarinic = muscarinic antagonist = **Parasympatholytic**

Sympathetic nervous system

- Mimic norepinephrine/epinephrine = adrenergic = adrenergic agonist = **Sympathomimetic**
- Block norepinephrine/epinephrine = anti-adrenergic = adrenergic antagonist = **Sympatholytic**

ANS Neurotransmitters

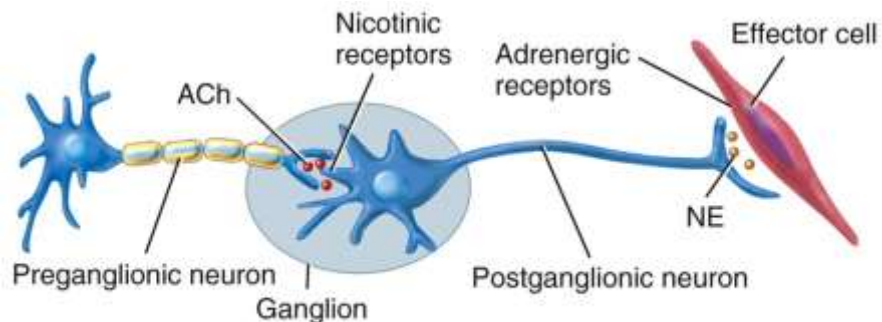
Neurons and Neurotransmitters of the Parasympathetic Nervous System	
Preganglionic	Postganglionic
Cell body in brain or spinal cord	Cell body in intramural ganglion
Acetylcholine (ACh)	Acetylcholine (ACh)



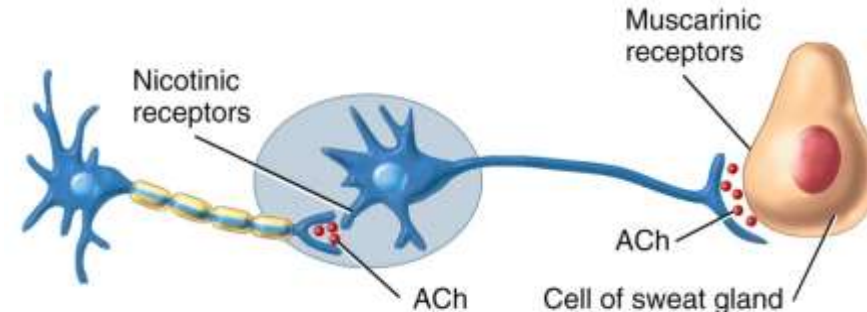
(c) Parasympathetic division

ANS Neurotransmitters

Neurons and Neurotransmitters of the Sympathetic Nervous System	
Preganglionic	Postganglionic
Cell body in lateral horn of spinal cord	Cell body in sympathetic chain ganglion
Acetylcholine (ACh)	(norepinephrine, NE) except sweat glands (ACh)



(a) Sympathetic division—innervation to most effector tissues

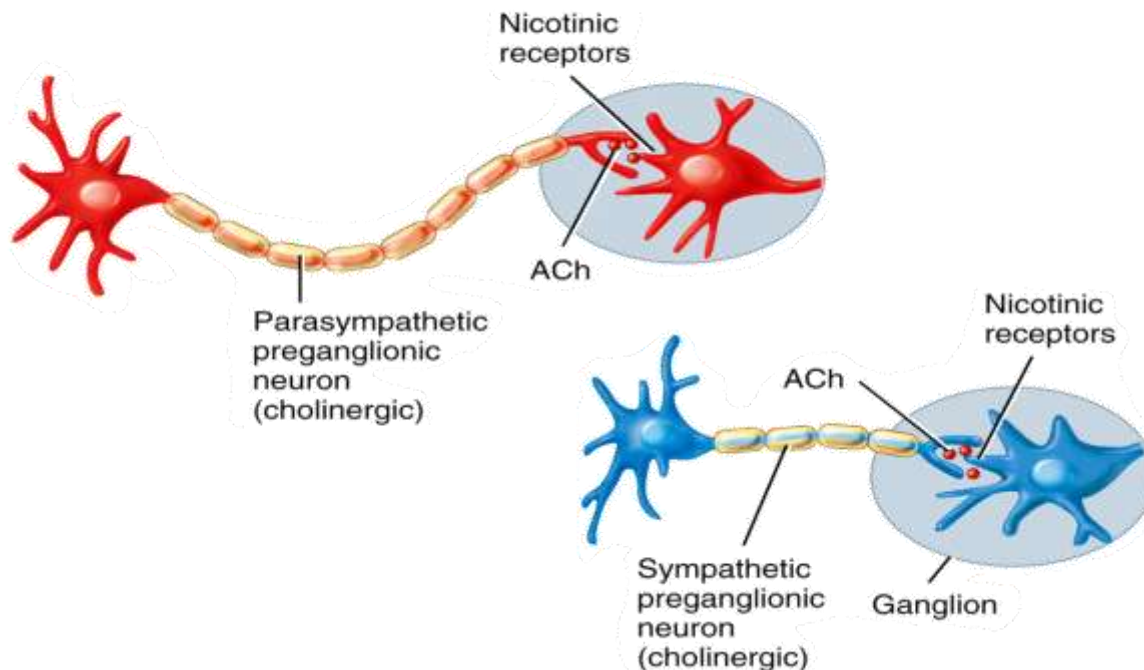


(b) Sympathetic division—innervation to most sweat glands

ANS Receptors & Actions

Cholinergic Receptors

- bind acetylcholine, ACh
 - **Nicotinic** – usually excitatory & quick
 - All post-ganglionic neurons



- **Muscarinic** – usually excitatory & slow

- Parasympathetic effector membranes

Interactions of the Autonomic Divisions

- Most visceral organs are innervated by both sympathetic and parasympathetic fibers
- This results in dynamic antagonisms that precisely control visceral activity
- Sympathetic fibers increase heart and respiratory rates, and inhibit digestion and elimination
- Parasympathetic fibers decrease heart and respiratory rates, and allow for digestion and the discarding of wastes

Sympathetic Tone

- The sympathetic division controls blood pressure and keeps the blood vessels in a continual state of partial constriction
- This sympathetic tone (vasomotor tone):
 - Constricts blood vessels and causes blood pressure to rise as needed
 - Prompts vessels to dilate if blood pressure is to be decreased
- Alpha-blocker drugs interfere with vasomotor fibers and are used to treat hypertension

Parasympathetic Tone

- Parasympathetic tone:
 - Slows the heart
 - Dictates normal activity levels of the digestive and urinary systems
- The sympathetic division can override these effects during times of stress
- Drugs that block parasympathetic responses increase heart rate and block fecal and urinary retention

Cooperative Effects

- ANS cooperation is best seen in control of the external genitalia
- Parasympathetic fibers cause vasodilation and are responsible for erection of the penis and clitoris
- Sympathetic fibers cause ejaculation of semen in males and reflex contraction of a female's vagina

Unique Roles of the Sympathetic Division

- **Regulates functions not subject to parasympathetic influence:**
 - Adrenal medulla (adrenaline release)
 - Sweat glands (cholinergic sympathetic)
 - Arrector pili muscles (piloerection)
 - Most blood vessels
 - Kidneys (renin release)
- **Thermoregulation:**
 - Heat → vasodilation, sweating
 - Cold → vasoconstriction
- **Metabolic Effects:**
 - Increases metabolic rate
 - Raises blood glucose (glycogenolysis)
 - Mobilizes fat (lipolysis)

Localized vs Diffuse Effects

Feature	Parasympathetic	Sympathetic
Duration	Short-lived	Long-lasting
Distribution	Highly localized	Diffuse, widespread
Reason	ACh rapidly hydrolyzed; single fibers innervate specific organs	NA inactivated slowly; interconnections in ganglia; adrenal release

LEARNING OUTCOMES

Students must be able to comprehend:

- Effects mediated by the autonomic nervous system
- The receptors that mediate the effects of the autonomic nervous system and their signal transduction mechanisms
- How drugs influence the autonomic nervous system

END

Thanks for listening